06 22313 Computational Tools for Modelling and Analysis

Dimensionality Reduction

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Introduction

The Curse of Dimensionality

Principal Component Analysis

Using the Principal Components

Alternatives to PCA

Outline

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The Curse of Dimensionality

Principal Component Analysis
- Using the Principal Components

Alternatives to PCA

Introduction

- Many types of data are *multivariate*:
  - Fluorescence imaging at multiple wavelengths.
  - Mass spectrometry data.
  - DNA Microarray data.
- Typically these datasets can have hundreds to millions of samples, of tens to hundreds of thousands of variables
- Hard to visualise effectively
- Two challenging cases:
  1) Large numbers of samples
  2) Large numbers of variables
- Focus here is on 2): high-dimensional data

Problems with high-dimensional data

- Hard to visualise > 3d
  - Tricks involving colours and shapes
  - Spider diagrams
- Computational cost.
- **Geometric intuition fails disastrously**
- Consider two "hyper"-spheres, of radius r and $r - \delta$
- Volume of larger sphere $V_1 \; r^d$
- Volume of smaller sphere $V_2 \; r^d - (r - \delta)^d$
- Ratio of volumes:

$$\lim_{d \to \infty} \frac{r^d - (r - \delta)^d}{r^d} = 1 \tag{1}$$

- Volume concentrates around the edge!

The Curse of Dimensionality

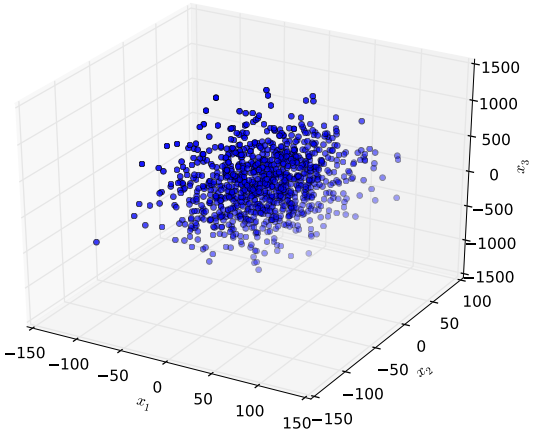
- Similar problems with distances in high dimensions
- Consider the minimum and maximum distances between pairs of points
- It can be proven that

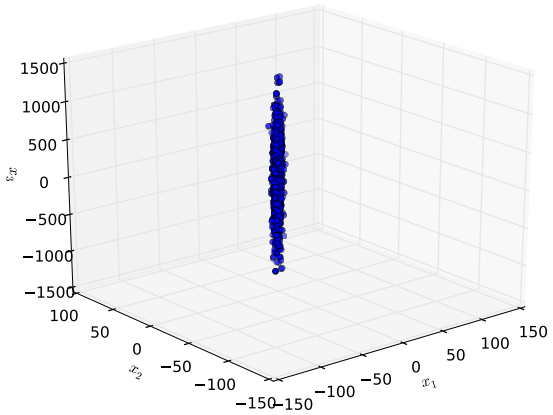
$$\lim_{d \to \infty} E\left(\frac{\text{dist}_{\text{max}}(d) - \text{dist}_{\text{min}}(d)}{\text{dist}_{\text{min}}(d)}\right) \to 0 \qquad (2)$$

- That is, all pairwise distances converge
- How on earth do we do clustering?!

Overcoming the curse

- High-dimensional data is frequently "redundant"
- Some dimensions will only be "noise"
- Variables may be very strongly correlated
- Often possible to find an alternative coordinate system that exposes this





Principal Component Analysis

- PCA allows us to find a **transformation** that removes redundant dimensions in a dataset
- Matches the coordinate system to the shape of the data such that:
  - The first coordinate is aligned with the direction of greatest variation in the data.
  - The second coordinate is perpendicular to the first coordinate and aligned with the direction of greatest variation under that constraint,
  - Subsequent coordinates are perpendicular to all that have already been defined, aligned with the direction of greatest variation under that constraint.

Performing PCA

- Covariance matrix, $\mathbf{\Sigma}$, describes the way different variables "co-vary":

$$\Sigma_{ij} = \frac{1}{N-1} \sum_{n=1}^{N} \left(x_i^{(n)} - \bar{x}_i \right) \left(x_j^{(n)} - \bar{x}_j \right)$$

  [$\bar{x}_i$ is the mean value of x_i.]
- Easy to compute $\mathbf{\Sigma}$ if we organise our dataset in a sensible way:
  - Construct matrix $\mathbf{X}$ with components $X_{ij} = x_i^{(j)}$
  - Each *row* of $\mathbf{X}$ corresponds to a variable
  - Each *column* corresponds to a sample
  - Compute the mean of every row and tile the resulting column vector side-by-side N times $\rightarrow \bar{\mathbf{X}}$.
  - Then, we form the covariance matrix as

$$\mathbf{\Sigma} = \frac{1}{N-1} \left(\mathbf{X} - \bar{\mathbf{X}} \right) \left(\mathbf{X} - \bar{\mathbf{X}} \right)^{\mathrm{T}} = \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathrm{T}} \qquad (3)$$

Performing PCA

- We will now find coordinate vectors that align with the covariance structure
- Define matrix $\mathbf{T} = \begin{pmatrix} \mathbf{t}_1 & \mathbf{t}_2 & \dots \mathbf{t}_M \end{pmatrix}$
- Columns of $\mathbf{T}$ (the vectors $\mathbf{t}$) represent form a new coordinate system (basis) for the space.
- Normalised such that $\mathbf{T}^{\mathrm{T}}\mathbf{T} = \mathbf{I} \leftrightarrow \mathbf{t}_i \cdot \mathbf{t}_j = \delta_{ij}$.
- First we find the direction of greatest variance
- Find the vector $\mathbf{t}_1$ that maximises the variance of the projection of the data.

$$\mathbf{t}_1 = \arg\max_{\mathbf{t}} \mathrm{var}(\mathbf{t}^{\mathrm{T}}\mathbf{X})$$

- Now $\mathrm{var}(\mathbf{t}^{\mathrm{T}}\mathbf{X}) = \mathbf{t}^{\mathrm{T}}\tilde{\mathbf{X}}\tilde{\mathbf{X}}^{\mathrm{T}}\mathbf{t} = \mathbf{t}^{\mathrm{T}}\mathbf{\Sigma}\mathbf{t}$ ignoring the pre-factor $1/(N-1)$
- Find $\mathbf{t} = \mathbf{t}_1$ that maximises this subject to $\mathbf{t}^{\mathrm{T}}\mathbf{t} = 1$

Performing PCA

- Mathematical "trick": Lagrange multiplier
- Subtract a "multiple of zero":

$$\mathbf{t}_1 = \arg\max_{\mathbf{t}} \left[\mathbf{t}^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{t} - \lambda_1(\mathbf{t}^{\mathrm{T}}\mathbf{t} - 1)\right] \tag{4}$$

- Differentiate this with respect to λ and set to zero to maximise

$$(\boldsymbol{\Sigma} - \lambda_1\mathbf{I})\,\mathbf{t}_1 = 0 \quad \rightarrow \quad \boldsymbol{\Sigma}\mathbf{t}_1 = \lambda_1\mathbf{t}_1 \tag{5}$$

- Solve "*eigenvalue*" problem for largest λ_1 and accompanying $\mathbf{t}_1$
- $\mathbf{t}_1$ is direction of greatest variation
- $\mathbf{t}_1^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{t}_1 = \lambda_1\mathbf{t}_1^{\mathrm{T}}\mathbf{t}_1 = \lambda_1$ is variance along that direction
- Next-greatest direction of variation is given by the second-largest eigenvalue/vector pair...

The principal components of a dataset are the eigenvectors of the covariance metric, ordered by eigenvalue.

- Once PCs have been found we transform the zero-mean data into the new coordinate system

$$\mathbf{X}' = \mathbf{T}^{\mathrm{T}}\tilde{\mathbf{X}} \tag{6}$$

- $\sqrt{(}\lambda_i)\mathbf{t}_i$ are called the **loadings**.
- Columns of $\mathbf{X}'$ are the **scores**: data projected onto the PC axes.
- Note that $\mathbf{\Sigma} = \mathbf{TST}^{\mathrm{T}}$ where $\mathbf{S} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.

Reducing dimensionality with PCA

- Projection of data onto loadings allows us to remove non-varying dimensions.
- Useful to consider the "total variance" $V_{\mathrm{T}} = \sum_i \lambda_i$.
- Compute proportion of total variance along each PC by computing $\lambda_i / V_{\mathrm{T}}$.
- Determine how many PCs describe the majority of the variance.
- Decide which ones to remove
- Set components of data in those dimensions to zero or equivalently, remove the corresponding columns form $\mathbf{T}$.

PCA Workflow

The application of PCA to a dataset will thus typically follow the following workflow:

1) Form the zero-mean matrix $\tilde{\mathbf{X}}$.
2) Calculate the covariance matrix $\mathbf{\Sigma}$
3) Calculate the eigenvalues/vectors of $\mathbf{\Sigma}$
4) Sort the eigenvalues/eigenvectors from largest to smallest by eigenvalue.
5) Choose a proportion of variance to retain and discard the eigenvalues/vectors for non-varying dimensions.
6) Compute the scores by projecting the data onto the new coordinates.

When to use PCA

- Often as part of the pre-processing workflow
- Relationships between points are preserved
- Dimensionality is reduced so distances now a useful discriminator
- Common to use PCA to reduce data size and dimensionality prior to clustering

Data Normalisation and Variable Scaling

- In some multivariate datasets, variables are of a different type or on a different scale
- Useful to consider mapping all variables onto a common scale to avoid bias
- Most common: rescale all variables to have unit variance.
- Take care when using of PCA in such a situation.

Random Projection

- Covariance matrix is computationally very expensive
- Have to pull some tricks for very large numbers of variables
- A deceptively simple but powerful alternative is the random projection
- Coordinate system is chosen at random
- Given a dataset $\mathbf{X}_{m\times n}$ of m variable from n samples, form a random matrix $\mathbf{R}_{m\times k}$ where each column is a normalised random vector:

$$\mathbf{R} = \begin{pmatrix} \mathbf{r}^{(1)} & \mathbf{r}^{(2)} & \dots & \mathbf{r}^{(k)} \end{pmatrix} \tag{7}$$

  where $\mathbf{r}^{(i)^{\mathrm{T}}}\mathbf{r}^{(i)} = 1$.
- Then in the new coordinates the scores are

$$\mathbf{X}' = \mathbf{R}^{\mathrm{T}}\mathbf{X}, \tag{8}$$

- Unlike PCA, there is no natural metric (variance) to decide how many projections are necessary.

Random Projection

- Surprisingly effective for dimensionality reduction
- Key mathematical result: Johnson-Lindenstrauss lemma
- Related to "concentration of measure" phenomena.
- Popular for several important reasons.
  - They are very inexpensive to compute.
  - For very high-dimensional data, random vectors are very likely to be orthogonal.
  - It can be shown that they preserve certain important distance metricss.
- Often worth trying due to the extreme low cost

Non-linear dimensionality reduction

- PCA and random projections are purely linear transformations of the data
- Do not change the "shape" of a cloud of datapointsF
- Effective for finding and removing globally non-varying directions in a dataset
- Not suited for finding finer structure in a dataset
- Popular alternative are:
  - Self-organising maps
  - Curvilinear component analysis
  - Latent variable models
  - t-distributed stochastic neighbour embedding (t-SNE – my personal favourite).
- Often use linear methods as pre-processing to eliminate globally redundant dimensions
- Then use a non-linear method or clustering to sort out the subleties.